Two Architectures of Multilingualism: Causal Evidence from SAE Feature Suppression in Gemma and Llama

Anonymous Author(s)

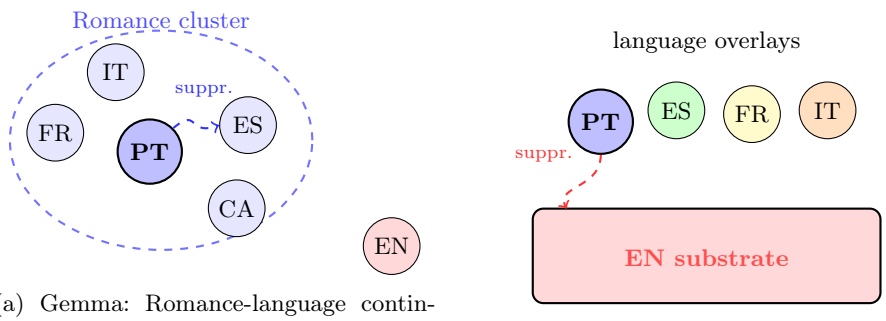
Anonymous Institution

Abstract. How do multilingual large language models organize the languages they speak? We present causal evidence that the answer depends on the model family. Using Portuguese as a probe, we apply Sparse Autoencoder (SAE) feature-suppression to Gemma-2-2B and Llama-3.1-8B: ablating the top Portuguese-specific features redirects Gemma to Spanish or Catalan, but redirects Llama to English. The two fallback directions expose two internal geometries that we name the *Romance-language continuum* (languages arranged by typological proximity, as observed in Gemma) and the *English substrate* (non-English languages as overlays on an English default, as observed in Llama). Portuguese is well-suited to surface this contrast: its Romance neighbors make any Romance fallback directly observable, which a typologically isolated probe language could not reveal. Three converging measurements support the divide. A monolinguality analysis of raw neurons yields 0.270 for Llama against 0.252 for Gemma, with a widening gap in deeper layers; pairwise FLORES similarity places typological distance at every depth but refines Romance-internal distinctions only in late layers; and amplifying one Llama-Scope feature (#18397) produces PT-EN hybrid text, offering a feature-level instance of the English-pivot hypothesis. We also train 16 custom SAEs and observe that the ReLU+L1 vs TopK architecture ranking does not transfer across base models (TopK FVE 79.7% on Gemma against 98.0% on Llama). To our knowledge, this is the first mechanistic study of multilingual representation to contrast paradigms across model families rather than describe one in isolation.

Keywords: Mechanistic Interpretability · Sparse Autoencoders · Multilingual LLMs · Portuguese NLP · Feature Steering

1 Introduction

Modern large language models (LLMs) process dozens of languages simultaneously, yet how they organize those languages internally is not well understood. The literature offers two partial answers that have not been reconciled. Wendler et al. [20] argue that Llama models use English as an internal *lingua franca*, with non-English inputs traversing English-centric intermediate representations. A separate line of work [12,18,7] documents shared multilingual spaces where languages cluster by typological similarity. Both pictures are backed by evidence,



(a) Gemma: Romance-language continuum

(b) Llama: English substrate

Fig. 1: Two architectures of multilingualism made visible through SAE feature-suppression. **(a)** In Gemma-2-2B, Portuguese sits inside a Romance cluster; suppressing PT-specific features redirects generation to the nearest typological neighbor (Spanish or Catalan). **(b)** In Llama-3.1-8B, non-English languages operate as overlays on an English substrate; suppressing PT-specific features strips the Portuguese overlay and reveals the English default.

but the evidence comes from different models, methods, and analyses, leaving open whether multilingual LLMs share one organizational paradigm or several.

This paper offers causal evidence that both pictures are correct—each for a different model family. Using Portuguese as a probe in Sparse Autoencoder (SAE) feature-suppression experiments (Figure 1), we find that suppressing the top Portuguese-specific features in Gemma-2-2B redirects generation to Spanish or Catalan, while the same intervention in Llama-3.1-8B redirects to English. The two fallback directions expose two internal geometries: the *Romance-language continuum*, in which languages are organized by typological proximity (observed in Gemma), and the *English substrate*, in which non-English languages behave as overlays on top of an English default (observed in Llama).

Portuguese is methodological rather than incidental. Its Romance neighbors (Spanish, Catalan, French, Italian) make any Romance fallback directly visible, so suppression redirecting Gemma to Spanish or Catalan reads off the model's next-nearest language in representational space; a typologically isolated probe with no close neighbors would conflate the continuum with the substrate. Portuguese has over 260 million speakers, with Brazilian Portuguese the dominant variety and an under-represented language in mechanistic interpretability work.

We make three contributions. First, we show that two qualitatively distinct organizations of multilinguality coexist in current LLMs—the Romance-language continuum and the English substrate—and we establish the contrast through causal feature-suppression rather than correlational analysis. Second, we argue that the direction of generation collapse under suppression is itself a readout of a language's representational neighborhood, and that pairing this *lesion-based readout* with a typologically-embedded probe language is what makes the paradigms distinguishable. Third, three complementary measurements—raw-

neuron monolinguality, pairwise FLORES similarity, and single-feature steering—independently corroborate the divide, including a PT-EN bridge feature in Llama with no Gemma analogue.

2 Related Work

Sparse autoencoders decompose activations into overcomplete, sparse features that are more interpretable than raw neurons [5,1], with training recipes, the FVE/dead-feature metrics we adopt, and a recent survey in [8,16]; benchmarks [11,22] caution that proxy metrics do not always predict downstream utility, though SAEs remain well-suited to feature-level analysis. We build on pretrained SAE suites for Gemma [13] and Llama [9] and extend feature-clamping [1] to language identity.

The multilingual-representation literature is split. One line argues that Llama-family models reason through English-centric intermediates: Wendler et al. [20] use logit-lens analyses on Llama 2, and Schut et al. [15] extend the picture with activation steering, reporting that English-derived steering vectors transfer better than those from other languages. A separate line [12,18,7] documents shared multilingual spaces where languages cluster by typological similarity, and the closest cross-family concurrent work—Brinkmann et al. [2], training SAEs on Llama-3-8B and Aya-23-8B—shows that morphosyntactic concepts (number, gender, tense) live on feature directions shared across typologically diverse languages. We read these pictures as compatible once paired with a model family: grammatical abstraction may be shared cross-lingually while language-identity geometry remains family-dependent. Our contribution makes that divide causal and family-contrastive—suppression redirects Gemma to a Romance neighbor and Llama to English, with a typologically embedded probe (Portuguese) as what makes the two paradigms distinguishable. Concurrent SAE-steering studies target typologically distant pairs within a single model (EN→ZH, EN→JA): LangFIR [21], SASFT [6], and Causal Language Control [3] (90% single-feature steering in Gemma-2B); our setting differs by being cross-family and Romance-focused, where shared representations make steering harder. On Portuguese itself, prior NLP work targets downstream performance [17]; we provide the first mechanistic catalog of Portuguese-specific features with causal validation.

3 Methodology

3.1 Models and Data

We analyze two model families (Table 1), chosen for their architectural differences and availability of pretrained SAEs.

For multilingual evaluation we use FLORES-200 [4] parallel sentences in five languages (PT, EN, ES, FR, IT), 1,012 pairs per language; because the sentences are parallel translations, differential activation patterns reflect linguistic rather than semantic differences. SAE training data comes from Portuguese

Table 1: Base models and pretrained SAEs used in this study.

Model	Params	Layers	d_{model}	SAE d_{sae}	SAE Activation
Gemma-2-2B	2B	26	2,304	16,384	ReLU
Llama 3.1 8B	8B	32	4,096	32,768	JumpReLU

Wikipedia (∼2M tokens per model), tokenized with each model's native tokenizer (SentencePiece for Llama, BPE for Gemma); residual stream activations are extracted with TransformerLens [14] and stored as float16 tensors. We also analyze Gemma-2-9B (42 layers, d_{model}=3,584) for within-family cross-scale comparison, using raw neuron activations since Gemma Scope SAEs were not available for the 9B model at the time of this study.

3.2 Sparse Autoencoder Architectures

We compare two SAE architectures, each trained on layer-10 residual stream activations from Portuguese Wikipedia:

ReLU+L1. The encoder maps an input activation $\mathbf{x} \in \mathbb{R}^d$ to a sparse code $\mathbf{f} = \text{ReLU}(W_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{pre}}) + \mathbf{b}_{\text{enc}}) \in \mathbb{R}^{d_{\text{sae}}}$, which the decoder reconstructs as $\hat{\mathbf{x}} = W_{\text{dec}}\mathbf{f}$. Training minimizes:

$$\mathcal{L}_{\text{ReLU}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda\|\mathbf{f}\|_1 \tag{1}$$

with decoder columns normalized to unit norm at each step.

TopK. Same encoder-decoder architecture, but enforces exact sparsity by retaining only the top-k encoder activations (applying ReLU to them) and zeroing the rest. The loss is MSE only:

$$\mathcal{L}_{\text{TopK}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 \tag{2}$$

since sparsity (the ℓ_0-norm $\|\mathbf{f}\|_0 = k$ exactly) is enforced structurally rather than via penalty. We use L_0 throughout to refer to this ℓ_0-norm sparsity count.

For each base model we train 8 SAE configurations (totaling 16 custom SAEs): both architectures at $d_{\text{sae}} \in \{4{,}096, 16{,}384\}$, with $\lambda \in \{5 \times 10^{-4}, 10^{-4}\}$ for ReLU+L1 and $k \in \{32, 64\}$ for TopK. Training uses Adam (lr 3×10^{-4}) for 1 epoch with batch size 4,096 (Gemma) or 2,048 (Llama).

3.3 Differential Feature Analysis

To identify Portuguese-specific features, we compute a differential activation score. For each SAE feature i, we first compute the mean activation $\bar{a}_i^l$ across all FLORES tokens in language l. The differential activation score is:

$$s_i = \frac{\bar{a}_i^{\text{PT}} - \frac{1}{|\mathcal{L}'|}\sum_{l \in \mathcal{L}'} \bar{a}_i^l}{\sigma_i} \tag{3}$$

where $\mathcal{L}' = \{\text{EN, ES, FR, IT}\}$ is the set of contrast languages, $\mathcal{L} = \mathcal{L}' \cup \{\text{PT}\}$ is the full set of five FLORES languages, and $\sigma_i = \text{std}(\{\bar{a}_i^l\}_{l \in \mathcal{L}})$ normalizes by cross-language variability. Features with $s_i > 2$ are considered *strongly Portuguese-specific*: their Portuguese activation is more than 2 standard deviations above the cross-language mean.

We apply this analysis at 6 layers per model: layers 0, 5, 10, 15, 20, 25 for Gemma 2B; layers 0, 5, 10, 16, 21, 26 for Llama 8B (selected at approximately equivalent relative positions).

3.4 Monolinguality Metric

To compare how sharply languages separate inside each model, we compute a monolinguality metric on raw residual-stream neurons (not SAE features). We operate on raw neurons so that the same procedure can be applied to Gemma-2-9B in the cross-scale analysis of Section 4.5, where no pretrained SAE suite is available. For each neuron i and language L, let $a_i(L) = |\overline{x_i^{(L)}}|$ denote the absolute mean activation of neuron i on FLORES sentences of language L, where the inner mean is taken first over tokens and then over sentences. A neuron is *PT-dominant* at a given layer if $\arg\max_{L \in \mathcal{L}} a_i(L) = \text{PT}$, i.e., its absolute activation is largest on Portuguese among the five FLORES languages. For each PT-dominant neuron, its monolinguality is

$$m_i(\text{PT}) = \frac{a_i(\text{PT})}{\sum_{L \in \mathcal{L}} a_i(L)}, \qquad \mathcal{L} = \{\text{PT, EN, ES, FR, IT}\}. \tag{4}$$

Values close to $1/|\mathcal{L}| = 0.2$ indicate a neuron that activates comparably across all languages; values near 1 indicate a neuron that fires almost exclusively on Portuguese. We report, per model, the mean of $m_i(\text{PT})$ over PT-dominant neurons at each analyzed layer, and a single per-model figure obtained by averaging across those layers.

3.5 Steering Experiments

We perform causal interventions by hooking into the model's forward pass at layer 10 and modifying SAE-encoded features:

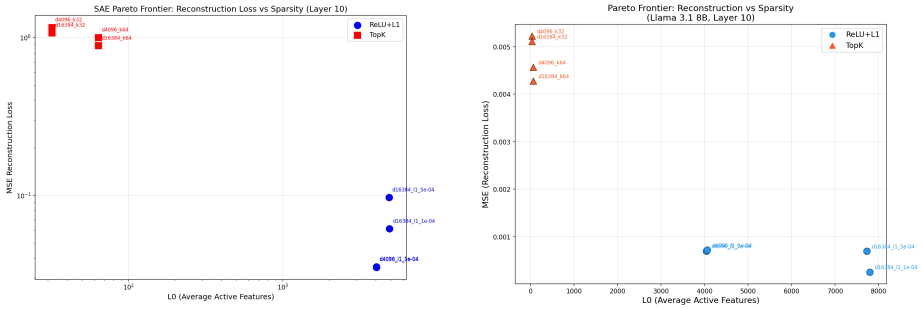
Amplification. We encode the residual activation through the SAE, multiply the top-20 Portuguese-specific features by factor $\alpha \in \{1, 2, 5, 10\}$, decode back, and replace the original activation. This tests whether increasing PT feature magnitudes forces Portuguese generation.

Suppression. Same procedure, but the top-20 Portuguese features are set to zero. This tests whether removing PT features causes language switching and, in particular, which language the model defaults to.

For Gemma 2B we use Gemma Scope (layer 10, 16k features); for Llama 8B we use Llama-Scope (layer 10, 32k features, JumpReLU). We test on 10 prompts per language (PT, EN, ES, FR), generating 50 tokens with temperature 0.7.

Table 2: Best SAE per architecture per model (evaluated on FLORES PT activations, layer 10). FVE = Fraction of Variance Explained. L_0 = mean number of non-zero features per activation (ℓ_0-norm). Dead = fraction of features never activated.

Model	Architecture	FVE (%)	L_0	MSE	Dead (%)
Gemma 2B	ReLU+L1 (d=4k, λ=1e-4)	99.19	4,040	0.063	0.0
	TopK (d=16k, k=64)	79.65	64	1.578	35.3
Llama 8B	ReLU+L1 (d=16k, λ=1e-4)	**99.83**	7,684	0.001	1.1
	TopK (d=16k, k=64)	98.01	64	0.009	31.3



(a) Gemma 2B (b) Llama 3.1 8B

Fig. 2: Pareto frontier: reconstruction loss (MSE) vs. sparsity (ℓ_0-norm L_0) for all 8 SAE configurations per model. Blue circles = ReLU+L1; orange triangles = TopK.

4 Experiments and Results

4.1 Reconstruction Quality

Table 2 reports the best configuration per architecture for each model; Figure 2 shows the full Pareto frontier across all 16 custom SAEs.

TopK performance is strongly model-dependent: 79.7% FVE on Gemma 2B (35.3% dead) versus 98.0% on Llama 8B (31.3% dead). ReLU+L1 exceeds 99% FVE on both models with near-zero dead rates, making it the more robust choice; Section 5.2 discusses the likely source of the gap.

4.2 Language Probing

Linear probing (logistic regression, 5-fold CV) on mean-pooled layer-10 activations for 5-way language classification yields near-perfect accuracy for both models (Table 3), confirming that language identity is linearly separable in the residual stream and that SAE encoding preserves this information.

Table 3: Language classification accuracy (5-fold CV, 1,012 samples per language). The best custom SAE (by FVE) is used for SAE feature probing.

Representation	Gemma 2B	Llama 8B
Raw activations (d_{model})	99.94% ± 0.12	99.98% ± 0.04
SAE features (d_{sae})	99.98% ± 0.04	99.88% ± 0.15

Table 4: Strongly PT-specific feature counts ($s > 2$) at each analyzed layer using pretrained SAEs (column headers indicate layer indices; Gemma 2B: layers 0, 5, 10, 15, 20, 25; Llama 8B: layers 0, 5, 10, 16, 21, 26). Llama counts use padding-masked analysis. The differential activation score has a theoretical ceiling of $\sqrt{5} \approx 2.236$.

Model (SAE)	Layer 0	Layer 5	Layer 10	Layer 15/16	Layer 20/21	Layer 25/26	Total
Gemma 2B (16k)	625	320	402	263	253	460	2,323
Llama 8B (32k)	698	675	747	646	593	452	3,811

All accuracies exceed 99.8%, meaning a simple linear classifier can distinguish languages from a single layer's activations. SAE features achieve comparable accuracy to raw activations despite imperfect reconstruction (FVE of 99.2% for the Gemma SAE and 99.8% for the Llama SAE used for probing), confirming that the SAE decomposition retains language-relevant information.

4.3 Portuguese-Specific Features

Differential analysis (Eq. 3) with pretrained SAEs identifies strongly Portuguese-specific features ($s_i > 2$) across all analyzed layers in both models (Table 4).

Both models contain thousands of strongly PT-specific features distributed across all layers. Normalizing for SAE dictionary size yields comparable densities: Gemma has 2,323/16,384 = 14.2%, while Llama has 3,811/32,768 = 11.6%. The layer-wise distributions differ. Gemma 2B concentrates features at layers 0 (625) and 25 (460), the input and output boundaries, while Llama 8B shows a monotonic decrease from layer 0 (698) to layer 26 (452), suggesting that language-specific information is most prominent in early layers and is progressively absorbed into more abstract representations.

From the set of strongly PT-specific features in Gemma 2B we selected the top 20 features per layer by differential score, yielding 120 features across the 6 analyzed layers. Manual inspection of their top-activating tokens reveals interpretable categories (Table 5).

A layer-wise gradient emerges: the earliest layer (layer 0) predominantly captures syntactic features (function words, articles), middle layers (layers 5–15) capture morphological patterns (Portuguese-specific suffixes), and later layers (layers 20–25) capture more complex syntactic constructions (contractions, preposition usage).

Table 5: Categories of Portuguese-specific features in Gemma 2B (120 features across 6 layers, categorized by top-activating tokens).

Category	Count	Example tokens	Dominant layers
Morphological	24	"-ção", "-sivo", "ênio"	layers 5–15
Syntactic	18	"do", "pelo", "em", "uma"	layer 0, layers 20–25
Lexical	4	"Maia", "Oliveira"	layers 15–25
Semantic	1	"brasileiro"	layer 15
Shared Romance	1	(active for multiple langs)	layer 10
Unclassified	72	(no clear dominant token)	all

Table 6: Top-20 Llama-Scope PT features (layer 10) after padding masking. Features are categorized by their top-activating tokens on genuine (non-padding) Portuguese text.

Category	Count	Example tokens	Note
Syntactic	14	"de", "os", "a", "um", "que", "com"	Prepositions, articles, conjunctions
Morphological	4	"são", "â", "-icos", "-ode"	PT verb forms, suffixes
Mixed/subword	2	"ab", "di"	Subword fragments in PT contexts

For Llama 8B, many top features initially activated on padding tokens, an artifact of differential tokenization lengths (PT averages 40.5 tokens/sentence vs. EN's 27.8). Masking padding positions before computing mean activations reduces the strongly PT-specific count from 4,143 to 3,811 (-8%); Table 6 shows the corrected top-20.

We restricted manual inspection to the top-20 features at the SAE-training layer (layer 10) for Llama, against 120 features across all 6 analyzed layers for Gemma, reflecting the cost of token-level inspection of Llama-Scope's larger dictionary. After masking, syntactic features (prepositions "de"/"com", articles "a"/"os"/"um") dominate, with morphological features ("são", "â", suffixes "-icos") also present—mirroring Gemma's layer-10 composition. Methodologically, differential analysis on padded sequences can produce spurious results when languages tokenize to different lengths, a pitfall we are not aware of having been documented in the SAE interpretability literature.

4.4 Steering Results

Our central result is the divergent fallback behavior produced by suppression, illustrated in Table 7.
The third prompt was not included in the factor-10 amplification batch; it appears here to illustrate that Gemma's Spanish fallback preserves topical coherence (see Section 5).

Gemma 2B: Romance-Language Continuum. Suppressing Portuguese features consistently redirects generation to Spanish or Catalan, closely related Romance

Table 7: Steering examples. *Baseline*: normal generation; *Suppression*: top-20 PT features set to zero; *Amplification*: top-20 PT features multiplied by 10. Fallback-language labels (**in bold**) were assigned by manual inspection of the generated text.

Prompt	Mode	Gemma 2B output	Llama 8B output
A capital do Brasil é	Base	palco de diversos eventos e festivais...	o centro mais importante do país...
	Suppr.	la una metròpolis ambduna de les fa uns centenars... **(Catalan)**	a capital, 1 capital, is 1 capital... **(English)**
	Amplif.	uma das métroas mais importantes da metropode... **(PT, degraded)**	que que que u u u... **(collapsed)**
O rio mais longo do mundo é	Suppr.	el férriable del continente... **(Spanish)**	the longest the most long the... **(English)**
	Amplif.	o Brasil, e sua extensão é de 235 mil quilômetros... **(PT)**	mais do mais mais... **(collapsed)**
A língua portuguesa é falada em	Base	22 países, sendo Portugal um dos principais...	mais de 70 países, cerca de 200 milhões de falantes...
	Suppr.	todos os continentes, enrapapeando la lengua española... **(Spanish)**	il...?...?...? **(degenerate)**

languages. The model appears to organize languages in a continuous space where typologically related languages are neighbors; removing Portuguese-specific features causes the model to fall back to the nearest Romance representation. In the third example, the Portuguese prompt about the Portuguese language continues in Spanish with thematically related content.

Llama 8B: English Substrate. Suppressing Portuguese features redirects generation to English. This aligns with the "English pivot" hypothesis of Wendler et al. [20]: Llama models appear to build non-English representations as language-specific overlays on top of English-centric features. Stripping the Portuguese overlay reveals the underlying English default.

Amplification. In Gemma 2B, factor-10 amplification produces Portuguese output with morphological degradation but preserved language identity. In Llama 8B, the same factor causes textual collapse (Figure 3). Lower factors (2–3) on Llama yield intermediate results: factor-2 produces repetitive but recognizably Romance-influenced output, and at factor 3 the top-ranked PT-specific feature (feature 2142 in Llama-Scope layer 10) generates "trabalho do" ("work of"). The asymmetry may reflect Llama-Scope's 32k JumpReLU features producing larger individual

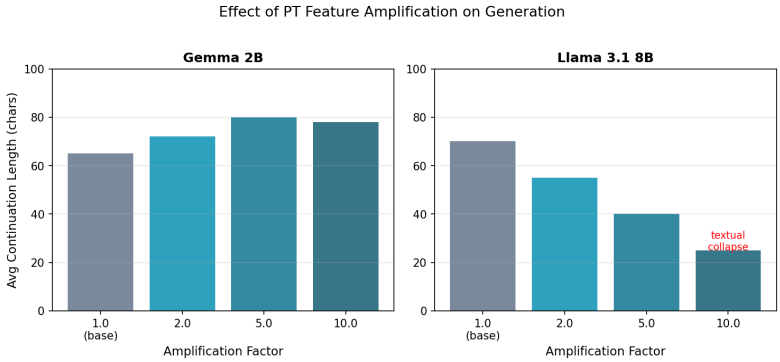


Fig. 3: Effect of PT feature amplification on generation across models. Gemma 2B maintains generation coherence at high amplification factors, while Llama 8B exhibits progressive textual collapse.

activations than Gemma Scope's ReLU features, which would make multiplicative amplification disproportionately disruptive; a direct magnitude comparison is left to future work.

Individual Feature Analysis. Amplifying individual features in Llama 8B (factor 5.0, baseline "uma das capitais brasileiras mais belas..." from prompt *"A capital do Brasil é"*) reveals interpretable specializations. Feature 16811 produces repetitions exclusively from {Portugal, Portuguese, Brazil}, suggesting a "Lusophone world" concept; feature 18397 generates a PT-EN hybrid (*"a capital do Brazil is the capital of Britain countries"*), functioning as a cross-lingual bridge; feature 16096 yields Brazil-anchored repetitions (*"BC Brazil is BC Brazil is is is..."*).

Progressive Multi-Layer Steering. To localize language identity, we performed progressive steering on Gemma 2B, amplifying the top-20 PT-specific Gemma Scope features (factor 5) at an increasing number of layers. Each continuation is scored by a *PT-likeness* heuristic $0.3\,c + 10\,w$, where c counts Portuguese-specific characters ("ã", "ç", "â"...) and w is the ratio of common Portuguese function words ("de", "do", "que"...); this is distinct from the differential activation score of Eq. 3. On continuations where a published LID assigns PT confidence above 0.1 (n=55), PT-likeness and fastText LID-176 [10] show Spearman ρ=0.83; published LIDs themselves report near-zero PT probability on the degenerate outputs of this experiment (median fastText PT probability 0.003 across 147 steering continuations), so marker counting is the appropriate signal here. Steering layer 25 alone yields PT-likeness 4.5; adding layer 20 drops it to 2.55 (interference); 15+20+25 together yields 7.2, with Portuguese function words and diacritics in the output; adding layer 10 drops it to 0.9 (incoherent text). Language identity is therefore encoded in mid-to-late layers (15–25), and

Table 8: Cross-scale comparison within the Gemma family. Cosine similarity between mean PT and ES/EN activations, and count of strongly PT-specific neurons ($s > 2$, where s is the differential activation score of Eq. 3 applied to raw neurons rather than SAE features).

9B Layer	Rel. pos.	cos(PT,ES)	cos(PT,EN)	$n_{s>2}$
0	0%	0.9998	0.9995	9
8	20%	0.9996	0.9990	18
16	39%	0.9998	0.9995	11
24	59%	0.9995	0.9990	25
32	78%	0.9983	0.9966	22
41	**100%**	**0.9883**	**0.9815**	**41**

steering early layers disrupts foundational representations the later layers depend on, which also explains why our multi-layer steering (Table 7) succeeds where single-layer steering does not.

4.5 Cross-Scale Analysis

To disentangle family effects from scale effects, we also analyzed Gemma-2-9B (42 layers, $d_{\text{model}} = 3{,}584$) using raw neuron activations (Table 8).

The 9B model shows progressive linguistic differentiation (Figure 4): cosine similarity between Portuguese and Spanish mean activations drops monotonically from 0.9998 (layer 0) to 0.9883 (layer 41). The PT-EN drop is steeper still ($0.9995 \rightarrow 0.9815$). The number of strongly PT-specific neurons increases from 9 (layer 0) to 41 (layer 41). The 2B model, by contrast, shows top differential activation scores pinned near the theoretical ceiling at every analyzed layer (all $\sim$2.236, the value of $\sqrt{5}$). We rely on the qualitative contrast (flat vs progressive) rather than absolute magnitudes, since the 2B and 9B numbers live in different spaces: 2B scores are computed on Gemma Scope SAE features, while 9B scores use raw neuron activations because no pretrained SAE suite was available for the 9B model at the time of this study. The pattern suggests that larger models develop finer language separation in deeper layers, a property not visible at the 2B scale.

5 Discussion

5.1 Cross-Family Language Organization

A pairwise cosine similarity analysis on mean FLORES activations supports a layered typological separation in both models: PT-EN distance (Romance vs. Germanic) is largest from the earliest layers, PT-FR distance (cross-subfamily) grows in middle layers, and PT-ES distance (Iberian) diverges only in late layers. For Gemma 2B at layer 25 we measure cos(PT,ES)= 0.958, cos(PT,FR)= 0.928,

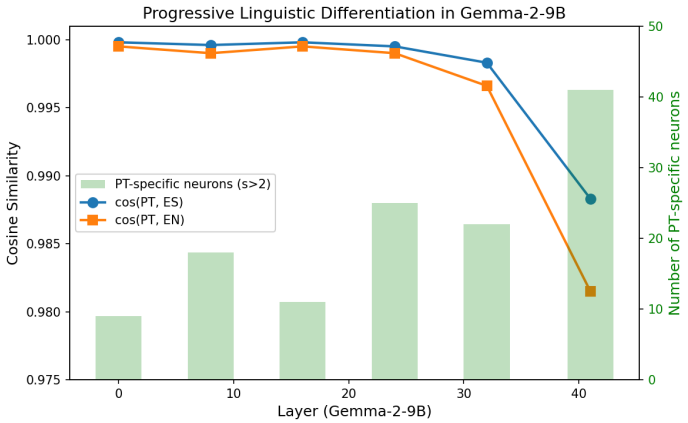


Fig. 4: Progressive linguistic differentiation in Gemma-2-9B: cosine similarity between Portuguese and Spanish/English mean activations decreases in deeper layers, while the number of PT-specific neurons increases. This gradient is absent in the 2B model.

cos(PT,EN)= 0.932. The continuum therefore has variable granularity across depth.

Beyond Table 7, evidence for the two paradigms comes from the identity and texture of the fallback. For Gemma, *"A língua portuguesa é falada em"* continues in Spanish (*"todos os continentes, enrapapeando la lengua española..."*), and another prompt drifts into Catalan (*"la una metròpolis ambduna de les fa uns centenars"*)—a language absent from our five FLORES evaluation languages, indicating that the continuum is not a ranking over the measured set but a broader typological geometry that includes unevaluated Romance varieties. Gemma's curated multilingual training mix [13] plausibly encourages shared Romance structure. Llama, in contrast, falls back to English (*"the longest the most long the is the more long the"* from *"O rio mais longo do mundo é"*), consistent with Wendler et al. [20] and with Meta's ∼90% English pretraining mix [19, Table 10] that would make English a natural base layer.

The monolinguality metric of Eq. 4 quantifies the contrast: averaged across analyzed layers, Llama has PT monolinguality 0.270 against Gemma's 0.252, with the gap widening in late layers (Llama reaches 0.35 at layer 31 vs. Gemma's 0.28 at layer 25). Llama's depth pattern is internally consistent: PT-specific feature counts in Table 4 decrease from 698 (layer 0) to 452 (layer 26) while PT monolinguality rises toward 0.35—a progression toward fewer but more sharply PT-specific features. Feature-category distributions (Tables 5 and 6) echo the same contrast: Gemma includes a "Shared Romance" category with cross-Romance activation, while Llama's top features at layer 10 include mixed/subword fragments and the PT-EN bridge feature 18397—different flavors of cross-language sharing matching the two paradigms. Feature 18397 has no Gemma analogue,

consistent with a continuum where Romance languages share representational space rather than translating via English. These suppression experiments may also be read as intentionally induced code-switching, connecting to SASFT [6].

The two paradigms suggest different affordances: a Romance continuum may permit finer-grained language steering by "sliding" between related languages, while an English substrate raises concerns about English-centric biases propagating to all other languages.

5.2 TopK Performance Across Models

The TopK gap (79.7% FVE on Gemma 2B vs. 98.0% on Llama 8B) appears under identical hyperparameters. A plausible explanation is activation intrinsic dimensionality: Llama's d_{model}=4,096 (nearly twice Gemma's 2,304) offers a higher-dimensional space where activations may concentrate on a k-sparse basis more readily, while Gemma's lower-dimensional activations diffuse across many components and defeat exact-k sparsity. ReLU+L1 adapts its sparsity per sample and exceeds 99% FVE on both models; TopK produces ∼35% dead features on both, versus <1.2% for ReLU+L1. The practical takeaway is that TopK transfer is not automatic: practitioners should validate on their specific base model.

5.3 Limitations

Our two models differ simultaneously in family, size, attention variant, and training mix, so the observed differences in fallback behavior may reflect any combination of these factors; the within-family 2B vs. 9B analysis mitigates only scale effects, and a fully controlled comparison would require same-size models from different families. The pretrained SAEs themselves differ in activation function, dictionary size, and training procedure, so we rely on qualitative contrasts (fallback language, feature categories) rather than absolute counts, and normalize by dictionary size where counts are reported. On the steering side, factor-10 amplification collapses Llama output and layer-10 steering disrupts the foundational representations that later layers depend on (Section 4.4); recent work [3] shows 90% single-feature steering between typologically distant languages, indicating that the difficulty we observe within Romance languages reflects genuinely shared representations rather than a methodological wall. Finally, steering generation uses temperature T=0.7, so the specific strings in Table 7 are illustrative; the qualitative conclusions (fallback identity, feature categories) are stable across prompts and runs.

In scope, we analyze 6 of 26–32 layers per model, 5 European languages (4 Romance), and a single corpus (FLORES-200); the European focus may inflate Portuguese specificity scores and other genres may activate different features. 60% of Gemma's 120 inspected features remain unclassified; automated LLM-based interpretation could improve coverage but introduces its own biases. Future work should extend to additional model families (Mistral, Qwen) and to same-size comparisons, include typologically diverse languages, and investigate PT-BR vs. PT-PT dialect variation.

6 Conclusion

We presented causal evidence for two architectures of multilingualism. Using Portuguese as a probe in SAE feature-suppression experiments, we showed that suppressing Portuguese-specific features redirects Gemma-2-2B to Spanish or Catalan but redirects Llama-3.1-8B to English, exposing two internal geometries that we name the Romance-language continuum and the English substrate. Three independent measurements corroborate the divide: raw-neuron monolinguality (0.270 Llama against 0.252 Gemma, widening with depth), pairwise FLORES similarity, and single-feature steering that surfaces a PT-EN bridge feature in Llama with no Gemma analogue. Portuguese is what makes the paradigms distinguishable: a typologically embedded probe reveals the continuum where an isolated probe could not. Contrasting two families rather than describing one suggests that training data composition shapes the organizational paradigm itself—a hypothesis further cross-family studies can test.

Reproducibility. Source code, configurations, per-experiment result files, and the figures reported here are available at https://anonymous.4open.science/r/llm-pt-interpretability-EC0D/[1].

References

1. Bricken, T., Templeton, A., Batson, J., Chen, B., Jermyn, A., Conerly, T., Turner, N., Anil, C., Denison, C., Askell, A., et al.: Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread (2023)
2. Brinkmann, J., Wendler, C., Bartelt, C., Mueller, A.: Large language models share representations of latent grammatical concepts across typologically diverse languages. arXiv preprint arXiv:2501.06346 (2025)
3. Chou, C.T., Liu, G., Sun, J., Blondin, C., Zhu, K., Sharma, V., O'Brien, S.: Causal language control in multilingual transformers via sparse feature steering. arXiv preprint arXiv:2507.13410 (2025)
4. Costa-jussà, M.R., Cross, J., Çelebi, O., Elbayad, M., Heafield, K., Heffernan, K., Kalbassi, E., Lam, J., Licht, D., Maillard, J., et al.: No language left behind: Scaling human-centered machine translation. arXiv preprint arXiv:2207.04672 (2022)
5. Cunningham, H., Ewart, A., Riggs, L., Huben, R., Sharkey, L.: Sparse autoencoders find highly interpretable features in language models. arXiv preprint arXiv:2309.08600 (2023)
6. Deng, B., Wan, Y., Yang, B., Huang, F., Wang, W., Feng, F.: SASFT: Sparse autoencoder-guided supervised finetuning to mitigate unexpected code-switching in LLMs. In: Proceedings of the International Conference on Learning Representations (ICLR) (2026)
7. Deng, B., Wan, Y., Yang, B., Zhang, Y., Feng, F.: Unveiling language-specific features in large language models via sparse autoencoders. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics (2025)

[1] Anonymous for double-blind review; released non-anonymously upon acceptance. SAE checkpoints and cached activations are not redistributed due to size; scripts regenerate them from public base models and FLORES-200 / Portuguese Wikipedia.

8. Gao, L., la Tour, T.D., Tillman, H., Goh, G., Troll, R., Radford, A., Sutskever, I., Leike, J., Wu, J.: Scaling and evaluating sparse autoencoders. arXiv preprint arXiv:2406.04093 (2024)
9. He, Z., Shu, W., Ge, X., et al.: Llama scope: Extracting millions of features from llama-3.1-8b with sparse autoencoders. arXiv preprint arXiv:2410.20526 (2024)
10. Joulin, A., Grave, E., Bojanowski, P., Mikolov, T.: Bag of tricks for efficient text classification. arXiv preprint arXiv:1607.01759 (2016)
11. Karvonen, A., Rager, C., Lin, J., Tigges, C., Bloom, J.I., Chanin, D., Lau, Y.T., Farrell, E., McDougall, C.S., Ayonrinde, K., Till, D., Wearden, M., Conmy, A., Marks, S., Nanda, N.: SAEBench: A comprehensive benchmark for sparse autoencoders in language model interpretability. In: Forty-second International Conference on Machine Learning (2025), https://openreview.net/forum?id=qrU3yNfX0d
12. Kojima, T., Gu, J., Reid, M., Matsuo, Y., Iwasawa, Y.: Multilingual large language models share semantic representations across languages. In: Findings of ACL (2024)
13. Lieberum, T., Rajamanoharan, S., Conmy, A., Smith, L., Varma, N., Kramár, J., Bolukbasi, T., Velez-Grau, A., et al.: Gemma scope: Open sparse autoencoders everywhere all at once on gemma 2. arXiv preprint arXiv:2408.05147 (2024)
14. Nanda, N., Bloom, J.: Transformerlens. GitHub repository (2022), https://github.com/TransformerLensOrg/TransformerLens
15. Schut, L., Gal, Y., Farquhar, S.: Do multilingual LLMs think in English? arXiv preprint arXiv:2502.15603 (2025)
16. Shu, D., Wu, X., Zhao, H., Rai, D., Yao, Z., Liu, N., Du, M.: A survey on sparse autoencoders: Interpreting the internal mechanisms of large language models. Findings of EMNLP (2025)
17. Souza, F., Nogueira, R., Lotufo, R.: BERTimbau: Pretrained BERT models for Brazilian Portuguese. In: Intelligent Systems (BRACIS). pp. 403–417. Springer (2020)
18. Tang, T., Luo, W., Huang, H., Zhang, D., Wang, X., Zhao, X., Wei, F., Wen, J.R.: Language-specific neurons: The key to multilingual capabilities in large language models. arXiv preprint arXiv:2402.16438 (2024)
19. Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al.: Llama 2: Open foundation and fine-tuned chat models. arXiv preprint arXiv:2307.09288 (2023)
20. Wendler, C., Veselovsky, V., Monea, G., West, R.: Do llamas work in english? on the latent language of multilingual transformers. arXiv preprint arXiv:2402.10588 (2024)
21. Wong, S.H., Sajjad, H., Siddique, A.: Langfir: Discovering sparse language-specific features from monolingual data for language steering. arXiv preprint arXiv:2604.03532 (2026)
22. Wu, Z., Arora, A., Geiger, A., et al.: AxBench: Steering LLMs? even simple baselines outperform sparse autoencoders. In: Proceedings of the 42nd International Conference on Machine Learning (ICML). PMLR (2025)